Poorv Patil

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EDUCATION

University of California, Berkeley Expected Graduation: May 2028

B.S. Electrical Engineering & Computer Sciences (EECS)

Relevant Coursework: Data Structures and Algorithms, Structure and Interpretation of Computer Programs, Signals and Dynamics, Discrete Math and Probability, Multivariable Calculus, Linear Algebra, Diff. Equations, Robotics & Automation

EXPERIENCE

Cognixion Inc. | *Software Engineering Intern* San Leandro, CA | Jun. 2024 – Aug. 2025

- Improved **top-3 next-phrase prediction accuracy by 15%** for SpeakProse 3 by rebuilding its PyTorch NLP pipeline around conversational context windows, evaluated on 10K+ user messages for faster assistive communication.
- Reduced **inference latency 35% (200ms to 130ms)** by caching context and batching scoring over 2K+ user sessions.

Algoverse AI | *Software Engineer Intern (LLM Security)* Fremont, CA | Feb. 2025 – Aug. 2026

- Achieved **87.3% detection accuracy** on cryptographic API misuse in LLM-generated code by building PyTorch activation-extraction and Sparse Autoencoder training over Gemma-2 9B representations and vulnerability features.
- Helped develop **CIPHER**, a reproducible benchmark and scoring pipeline spanning 450 prompts, 17 vulnerability categories, 7 LLMs, and 2,250 generations/model with line-level evidence attribution and schema-validated evaluation.

UCSB ECE Department | *Software Engineering Intern (Robotics)* Santa Barbara, CA | Jun. 2025 – Aug. 2025

- Built **C++/MATLAB planning and localization software** for a 3-UAV search-and-rescue system, integrating stochastic lost-person behavior, cellular-signal localization, and multiple search strategies across 1K+ Monte Carlo simulations.
- Implemented MATLAB evaluation infrastructure to stress-test missions under wind, network loss, and fleet constraints, automating benchmarking against Civil Air Patrol baselines; sustained **70% detection probability** on long searches.

University of Oxford | *Machine Learning Intern (Signal Processing)* Oxford, U.K. | Jun. 2023 – Nov. 2025

- Built an **XGBoost pipeline** for cardiac abnormality detection on 1,112 phonocardiograms under Oxford postdoc Dr. Shadi Ghiasi, reaching **92% accuracy** on low-compute clinical hardware and beating the prior 81% benchmark by 11 points.
- Bandpass-filtered raw heart-sound audio and engineered 25 features, including a novel entropy-fluctuation metric; a manually selected 10-feature subset outperformed the full set, and SVM and KNN baselines both confirmed robustness.
- Published **first-author paper in IEEE Xplore**; youngest speaker at IEEE BIBE 2024, then presented to Nobel Laureates.

TECHNICAL PROJECTS

Autonomous Quadcopter Engineering | ***ROS, Gazebo, PyTorch, Pixhawk*** Fremont, CA | May 2026 – Present

- Built a custom **F450 quadcopter** from scratch around a Pixhawk flight controller, soldering ESC/PDB power connections and 3D-printing a **custom vibration-damping mount** to isolate the onboard sensors from motor vibration and noise.
- Developed ROS/Gazebo **simulation pipelines for autonomous UAV navigation**, integrating reinforcement learning with **Kalman filter-based state estimation** to evaluate trajectories across simulated missions.

CourseSync: Gradescope Deadline Calendar | ***FastAPI, Google Calendar API, OAuth 2.0*** Apr. 2026 – Present

- Built a **FastAPI backend** that synchronizes Gradescope assignment deadlines to **Google Calendar** as structured events.
- Implemented OAuth 2.0 access and PostgreSQL mappings to create, update, and deduplicate events as deadlines change.

TripSync: Collaborative Group Travel Planner | ***React, FastAPI, PostgreSQL, WebSockets*** Jun. 2026 – Present

- Built a full-stack group travel planner with a **FastAPI REST API**, **PostgreSQL persistence**, scheduled background jobs, and **Dockerized deployment** for shared itinerary updates, collaborative planning, and recurring data synchronization.

fairHealth: AI for Predatory Hospital Billing | ***React, GANs, NLP*** Ross, CA | Mar. 2024

- Placed **2nd of 20+ international teams** at Bull Hacks by building fairHealth, a React web app that parses hospital bills and benchmarks charges against regional pricing using a geolocation pricing API, GAN estimator, and NLP pipeline.

LEADERSHIP

ExpertiseNPO | *Co-Founder* Fremont, CA | Sep. 2023 – Present

- Co-founded a 501(c)(3) delivering free STEM/AI education to **800+ K-8 students** across 8 multi-week bootcamps; recruited and managed 28 expert instructors and raised $6.2K+ while partnering with Milpitas Unified School District.

SKILLS & AWARDS

Languages: Python, C++, Java, JavaScript, SQL, C, MATLAB, HTML/CSS

Libraries/Tools: AWS, Docker, PyTorch, scikit-learn, React, FastAPI, ROS, Gazebo, Git, NumPy, pandas, PostgreSQL

Awards: National Merit Scholarship Recipient, AIME Qualifier, 3x American Computer Science League Finalist